



Vavuniya Campus of the University of Jaffna

Fourth Examination in Information and Communication Technology - 2018

First Semester - September/October 2019

ICT4171 Practical for Bioinformatics and Computational Biology

Answer All Questions

Time Allowed : Two hours

Instructions

- Documents related to the questions are saved in the folder named *ICT4171_Final* and placed in **fleshare** drive. Save your files in a new folder named with your index number (E.g.: *IT10005*) in your Z drive.
1. The most common type of sequencing error occurs when a single nucleotide from a read is interpreted incorrectly. There is a collection of up to 1000 reads of equal length (at most 50 bp) in FASTA format. Some of these reads were generated with a single-nucleotide error. For each read *s* in the dataset, one of the following applies:
 - *s* will be assumed to be correctly sequenced if it appears in the dataset at least twice or as a reverse complement.
 - *s* will be assumed to be incorrect, if it appears in the dataset exactly once, and its Hamming distance is 1 with respect to exactly one correct read in the dataset or its reverse complement.

Write a program in Python to list all the corrections in the form of "old read-> new read". (Each correction must be a single symbol substitution, and you may return the corrections in **any order**.)

To be continued

The Input Dataset is

TCATC

TTCAT

TCATC

GATGA

TGAAA

GAGGA

TTTCA

ATCAA

TTGAT

TTTCC

The Output is

TTCAT->TTGAT

GAGGA->GATGA

TTTCC->TTTCA

Note: The Hamming distance between two strings having the same length is the minimum number of symbol substitutions required to transform one string into the other.

[60%]

2. Use the `Child_Variants.csv` dataset given in *ICT4171_Final* folder and write a script in R Language for each of the tasks given below:

- (a) Create a filtered version of the `child.variants` data that includes only rows where `MutantReadPercent` ≥ 70 .
- (b) Draw a histogram of the `MutantReadPercent` column of the child variants dataset.
- (c) Plot a boxplot of the `MutantReadPercent` values from both the original child variants and the same column from the filtered dataset (`MutantReadPercent` ≥ 70).
- (d) Find the frequency with which each of G, A, T and C bases were mutated, and Plot the results as a barplot with original names of the nitrogen bases.

[40%]